MIL-S-60619A(MU)
31 October 1972
SUPERSEDING
MIL-s-60619(Mu)
15 November 1966

MILITARY SPECIFICATION

SIGHT, REFLEX: M16 AND XM75

1. SCOPE

1.1. This specification covers two similar models of reflexsight. A reflex sight is designed to superimpose the reticle and the target on the same focal plans.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-8811	Chamber, Rain Testing
MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

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DRAWINGS

U.S. Army, Frankford Arsenal

F7597799	Sight, Optical: 7597799
F8565556	Pressure Tester-Two Station: 8565556
F11069411	Sight, Optical: 11069411

PACKAGING DATA SHEET

7597799	Packaging of Sight, Optical: 7597799
11069411	Packaging of Sight, Optical: 11069411

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Fabrication.- The Sight shall be manufactured in accordance with the applicable drawing and drawings pertaining thereto.

3.3 General Specifications.- The contractor shall be responsible for adherence to, and compliance with the following requirements of specifications MIL-O-13830 and MIL-F-13926.

3.3.1 Specification MIL-O-13830.

(a) Lenses

3.3.2 Specification MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surfaces finishes
(d) Part identification and marking
(e) Soldering electrical connections
(f) Workmanship

3.4 Environmental.

3.4.1 Pre-vibration collimation.- With the mounting surface of Adapter C7675759 positioned such that the width of the mounting surface is in a vertical plane, the line of sight through the center of the geometric axis of the projected reticle pattern shall be horizontal within plus or minus 1.0 mil and perpendicular to the mounting surface

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within plus or minus 1.0 roil. The position of the line of sight through the geometric axis of the projected reticle pattern shall be noted and recorded.

3.4.2 Operating temperatures. The Sight shall meet the operability requirements (rheostat knob and housing knob) while exposed to, and stabilized at, temperatures of -40°F and +150°F.

3.4.3 Storage temperature.- The Sight shall show no evidence of physical failure and shall meet the requirements of this specification at ambient temperatures 75 ± 15°F, after having been exposed and thermally stabilized at ambient temperatures of -80°F and +160°F.

3.4.4 Vibration.

3.4.4.1 Vibration (XM 75).- The sight shall not be damaged after being subjected to vibrations at a constant frequency of 30 CPS at an amplitude of 1/8 inch total excursion for periods of 10 minutes in each of three mutuals perpendicular directions (total 30 minutes) relative to its normal operation position. During vibration, there shall be no evidence of the reticle light and the launch envelope indicator light flickering.

3.4.4.2 Vibration (M 18).- The sight shall not be damaged after being subjected to vibrations at a constant frequency of 30 CPS at an amlplitude 1/8 inch total excursion for periods of 10 minutes in each of three mutual perpendicular directions (total 30 minutes) relative to its normal operating position. During vibration, there shall be no evidence of the reticle light flickering.

3.4.5 Rain (nonoperating).- After being exposed to a total of 1.50 inches of rain in a 10 minute period, with intermittent wind speeds up to 35 knots (to include a 1 inch fall in a 5 minute period that includes a 0.45 inch fall in a 1 minute period), there shall be no evidence of water penetration in the interior of the sight.

3.5 Post-vibration collimation.- The position of the geometric axis of the projected reticle pattern in relation to the optical line of sight shall be within 1,0 mil in elevation and 0.5 mil in azimuth of the readings recorded in 3.4.1 (pre-vibration) with the sight positioned as specified in 3.4.1 (pre-vibration).

3.6 Performance characteristics.

3.6.1 Inputs.- The sight shall perform as specified herein when furnished with the following externally separated input signals.

3.6.1.1 Reticle lamp power.- Electrical characteristics of the input power required to energize the reticle lamp shall be as follows:

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a.	Voltage:	10.5 to 13.5 volts direct current (vdc)
b.	Current available:	Not more than 120 milliamperes (ma) at 13.5 vdc

3.6.1.2 Launch envelope indicator light power (XM 75).- Electrical characteristics of the input power required to energize the launch envelope indicator light shall be as follows: .

a.	Voltage:	18 to 30 vdc
b.	Current available:	Not more than 250 ma at 30 vdc

3.6.2 Reticle image intensity.- The reticle image intensity, as it appears to the observer, shall be adjustable from completely off to a level which is clearly defined when viewed against skylight or against a background brightness of 300 plus or minus 60 foot candles.

3.6.3 Parallax.- The angular deviation due to parallax shall be not greater than 0.2 milliradian (mr) when targets at ranges greater than 1000 meters are viewed.

3.6.4 Image quality.- The image of the reticle shall be clear and uniform without any break or irregularities.

3.6.5 FOcUS.- The focus of the reticle shall not be significantly affected over eye positions varying between 3 and 8 inches from the reflex plate.

3.6.6 Alignment shift.- The alignment shift of the reticle pattern line-of-sight with respect to the mounting base shall be not greater than 1 mr over the environmental extremes and operating conditions specified herein that may occur between boresight checks.

3.6.7 Boresight adjustments.- The line of sight through the center of the projected reticle pattern shall be parallel to the-mounting surface plane and shall be adjustable over an angle of not less than 2 degrees of arc in any direction. This requirement shall be met with the sight positioned as specified in 3.4.1 (pre-vibration collimation).

3.7 Operability.

3.7.1 Rheostat knob operability.- The rheostat knob shall be operable throughout its entire range when exposed to the operating temperatures specified in 3.4.2.

3.7.2 Housing knob.- The housing knob shall be capable of ratating the light housing from the daylight position to the night position and

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back including locking and unlocking in the detent position when exposed to the operating temperatures specified in 3.4.2.

3.8 Cleaniness.-There shall be no dirt within the sight which adversely affects the image quality of the projected reticle. The sight shall meet the cleanliness and optical Requirements of MIL-O-13830.

3.9 Sealing.- The sight (body assembly) shall withstand an internal pressure of five (5) psi of dry nitrogen gas for one hour without showing a pressure drop in excess of 0.05 psi. Subsequent to the above, the interior atmosphere of the body assembly shall have a clew point of -25°F.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilityfor inspection.- Unless otherwise specified in the contract or purchase order, the suppliers is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification, Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three sights plus three each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment,processes, and procedures as used in regular production. All parts and materials, including packin and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of Product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2.1 Comportments and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP) In the absence of SQAPS, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 First article tests.- Unless otherwise specified, the requirements and tests of Table I shall be conducted on first articles only. In the event that the processes, materials or other changes in design or manufacture occur subsequent to first article approval, the first items so produced shall be submitted to the Government in accordance with 4.2 for approval prior to resumption of production.

TABLE I

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Temperature (Storage)	3.4.3	4.6.1.2
302. Rain	3.4.5	4.6.1.3

4.3.2.3 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables II and III herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

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TABLE II CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Pre-vibration collimation	3.4.1	4.6.1.4
102. Vibration	3.4.4	4.6.1.5
103. Post vibration collimation	3.5	4.6.1.6
104. Reticle image intensity	3.6.2	4.6.2.2
105. Parallax	3.6.3	4.6.2.3
106. Image quality	3.6.4	4.6.2.4
107. Focus	3.6.5	4.6.2.5
108. Alignment shift	3.6.6	4.6.2.7
109. Boresight adjustments	3.6.7	4.6.2.6
110. Operability - Rheostat and Housing knobs	3.7.1 3.7.2	4.6.3.1
111. Cleanliness	3.8	4.6.4
112. Sealing	3.9	4.6.1.7
MINOR: NONE		

NOTE: Unless otherwise specified, the above tests shall be performed at room ambient temperature (60°F to 90°F).

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- Ten sights shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

	CHARACTERISTIC		TEST PROCEDURE
401.	Fabrication	3.2	Applicable drawing visual
402.	General specification	3.3	MIL-F-13926 - visual
403.	Temperature (operating)	3.4.2	4.6.1.1
404.	Alignment Shift	3.6.6	4.6.2.7
405.	Operability - Rheostat and Housing knobs	3.7.1, 3.7.2	4.6.3.1

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4.4.1 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the “Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and “Test Facilities"requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2, the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6.

a. Vibration and temperature test equipment conforming to the “Test Facilities” requirements of MIL-F-13926.

b. A receiving device capable “of positioning the sight as specified in 3.4.1. The device shall be mounted on a surface plate and cross leveled within ten seconds.

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c. A target projector collimator with a clear aperture of at least 2.0 inches containing a reticle test pattern capable of measuring the optical requirements specified.

d. A calibrated mirror adapter utilized to auto-collimate the target projector collimator.

e. An auxiliary 3 power Telescope.

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Operating temperatures.-This test shall be performed as outlined in MIL-F-13926, Procedure I. The sight shall be tested for operability of the rheostat and housing knobs when thermally stabilized at the temperatures specified for compliance with 3.4.2.

4.6.1.2 Starage temperate.- This test shall be performed as outlined in MIL-F-13926, Procedure II The sight shall be exposed. and thermally stabilized at the extreme temperatures specified in 3.4.3.Upon completion of this test, the sight shall be visually examined for damage, then tested as specified in 3.4.3.

4.6,1.3 Rain test.- This test shall be performed as outlined in MIL-F-13926, Procedure I, except the duration, wind velocity and amount of rain fall shall be as specified in 3.4.5. The test equipment utilized shall comply With MIL-C-8811. Upon conclusion of this test, the sight shall be visually examined for compliance with 3.4.5.

4.6.1.4 Pre-vibration collimation.- The target projector collimator (4.5.3.c) shall be mounted on a surface plate and cross leveled within ten seconds. A calibrated mirror adapter (4.5.3.d) positioned on the receiving device (4.5.3.b) utilizing the sight's mating datum surface of the device. The reflecting surface of the mirror shall be positioned perpendicular to the collimator's line of sight. The target projector collimator shall be auto-collimated with the reflected image coincident with the reticle pattern of the target projector collimator. Remove the mirror adapter from the receiving device then mount the sight on the device positioned as specified in 3.4.1. Illuminate the target projector reticle and the sight retiicle. Position the auxiliary telescope about 5 inches from the reflex plate of the sight. Sighting through the telescope, observe that the line of sight of the center of the projected reticle pattern is within the limits specified in 3.4.1 in the horizontal and vertical planes as measured on the target pro-jector reticle. The position of the line of sight shall be noted and recorded.

4..6.1.5 Vibration.- The sight shall be positioned and secured on a vibration table with the line of sight of reflected reticle image in

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a horizontal plane. The reticle light shall be illuminated by adjusting the rheostat. The sight shall then be vibrated at the amplitude, frequency and duration specified in 3.4.4. During vibration, the illuminated lights shall be visually observed and there shall be no noticeable flickering. Upon completion of the vibration test, the sight shall be examined by visual and tactile means to detect loose or broken components.

4.6.1.6 Post-Vibration collimation.- Position the sight as outlines in 4.6.1.4. Illuminate the target projector reticle and the sight reticle. Observe the line of sight through the auxiliary telescope on the reflected sight reticle with respect to the target projector reticle. The measured collimation error shall be within the limits specified in 3.4.1 and 3.5.

4.6.1.7 Sealing.- This test shall be performed using testing equipment the accuracy of which shall conform to the accuracy depicted on Drawing F8565556. The fixture shall be set-up as outlined in set-up instructions shown on Drawing F8565556, Sheet 1. The gas used shall be dry nitrogen having a dew point of -25°F. After connecting the test fixture to the sight (body assembly), adjust testing pressure to the pressure specified in 3.9 and maintain the pressure for time specified. At the completion of this test, connect a dew point tester to the exit port and purge the sight until -25°F is recorded and seal the unit.

4.6.2 Performance.

4.6.2.1 Inputs.- The power applied to illuminate the reticle lamp and launch envelope indicator shall be as specified in 3.6.1.1 and 3.6.1.2.

4.6.2.2 Reticle image intensity.- Voltages between 10.5 and 13.5 vdc shall be applied to the reticle lamp in increments of 1 volt. Lamp intensity shall be adjusted to its maximum setting for each applied voltage and the projected reticle image shall be observed with the unaided eye against a background of 300 plus or minus 60 foot-candles. The reticle image shall be clearly defined under these conditions. For compliance with 3.6.2.

4.6.2.3 Parallax.- The sight shall be positioned as outlined in 4.6.1.4.Remove the auxiliary telescope. The reticle lamp shall be illuminated and the sight shall be adjusted to center its reticle on target projector reticle. The target shall be observed through the sight at a distance of 5.5 inches from the reflex plate with the observer’s head moving horizontally and vertically through the optical center line plus and minus 1 inch. Under these conditions, the projected reticle shall not fall outside the limits in 3.6.3 as measured on the target reticle.

4.6.2.4 Image quality.- The reticle lamp shall be illuminated and adjusted to its maximum intensity. The projected reticle shall

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be observed against a uniform background such as a frosted glass plate. The image quality shall be clear, and uniform without any break or Irregularities as specified in 3.6.4.

4.6.2.5 Focus.- The reticle lamp shall be illuminated and adjusted to its maximum intensity. The projected reticle shall be observed against a uniform background such as a frosted glass plate. The projected reticle shall then be observed at distances of 3 inches and 8 inches from the reflex plate, Focus of the reticle shall not be significantly affected as specified in 3.6.5.

4.6.2.6 Boresight adjustments.- The sight shall be positioned as outlined in 4.6.1.4, with its line of sight parallel so the line formed by the intersection of reference surface "A" and reference surface "B" (see figure 1.) Adjustments shall be made in azimuth and in eelevation. Full adjustment excersion shall be not less class specified in 3.6.7 in azimuth and in rlevation as on the target reticle.

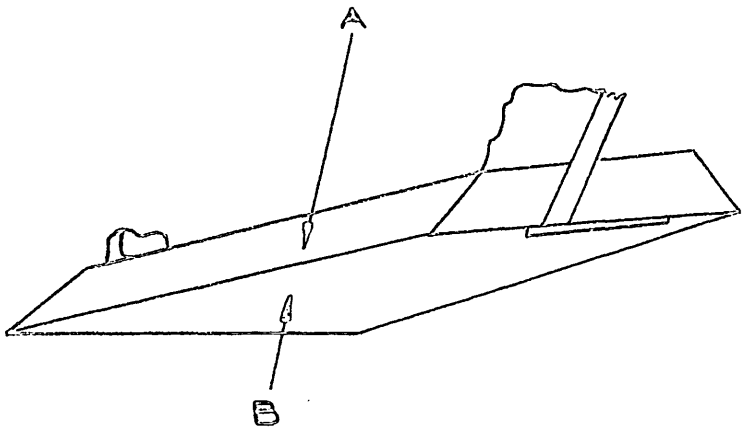


Figure 1 - Sight Mounting Surface

4.6.2,7 Alignment shift. - Perform this test at the extreme temperature operating conditions specified in 4.6.1.1 with the sight positioned and tested as outlined in 4.6.2.6 and 3.6.6. Observe and note any deviation of alignment that may occur during and after exposure to the extreme temperature, The alignment shift shall not exceed the limit specified in 3.6.6 as measured on the target reticle.

4.6.3 Operapility.

4.6.3.1 Rhestat knob and Housing knob.- With the sight exposed, to the extreme temperature and tested as outlined in 4.4.6.1.1 the

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rheostat knob and the housing knob shall operate without undue binding or freedom for compliance with 3.7.1 and 3.7.2.

4.6.4 Cleanliness.- The optical components of the sight shall be inspected for cleanliness in accordance with MIL-O-13830 for compliance with 3.8.

5. PREPARATION FOR DELIVERY

5.1 Preservation, Packaging, Packing, Marking- Preservations packaging and marking shall be in accordance with packaging data sheet 11069411 and packaging data sheet 7597799.

6. NOTES

6.1 Intended use.

6.1.2 XM75 sight is intended to be part of a fire unit to be used in a missile-launching mount assembly installed on a tracked cross-country vehicle.

6.1.3 M18 sight is intended to be used as a tracking aid for a (Cal .50 machine gun).

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging, and packing (see 5.1).
c. Applicable packaging data sheet number (see 5.1).
d. Applicable stock number.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A410

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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